

Automated Test Application Development Made Easy with ATEasy

Marvin Test Solutions' ATEasy is targeted at people engaged in the development and use of automated test equipment. Here we discuss its features, hardware and software specifications along with its workspace environment, modules and applications. A 30-day fully functional, free trial of ATEasy is included in the DVD accompanying this month's EFY Plus

EFY BUREAU

ATEasy is a test executive and a software development environment for test and measurement (T&M) applications. It contains all the tools required to develop test applications for automated test equipment (ATE) systems and instrument control applications.

ATE systems are used to perform testing on one or more electronic products called units under test (UUTs) such as components, boards and assemblies. Typically, these consist of a computer/controller, several test and measurement instruments and a test application designed to control system instruments in order to test the UUT.

The ATEasy integrated development environment (IDE) allows users to quickly develop automated test programs using a wide variety of hardware and software assets including PXI, PCI and GPIB as well as LabVIEW, C++ and .NET. ATEasy facilitates rapid application development but it is greatly focused on test development.

ATEasy has many built-in structures that are object-oriented and allow developers to focus on building a test without having to worry about certain detailed implementations such as log reporting.

The ATEasy test executive provides a highly intuitive graphical user interface (GUI) for executing ATEasy test programs. The ready-to-use GUI adapts to the test environment that it is linked to. It also frees the test developer from detailed implementation that he would typically need to develop from scratch when using other test platforms.

Why use ATEasy for automated test systems?

An automated test system, also referred to as automated test equipment (ATE), is a collection of instruments under computer control performing automated test functions. Fig. 1 shows the typical configuration of an ATE system. A computer provides control over test and measurement instruments by using hardware interfaces. The instruments, such as measurement, stimulus, switching, power and digital instruments, are connected to the UUT through an adaptor.

The most common computing tool used in ATE applications is the PC. Due to its relatively low-cost computing power and availability of

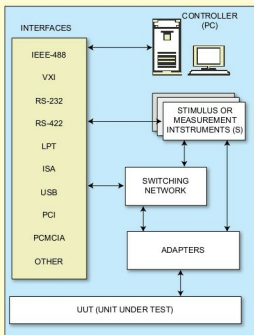


Fig. 1: Typical configuration of an ATE system

hardware interfaces as well as computer programs, the PC has become the de-facto standard of the test industry. The PC supports numerous interfaces for controlling test instruments. These interfaces include IEEE-488 (GPIB), VXI, ISA bus, PXI/PCI bus, LXI/TCP-IP, serial communication such as RS-232/422/485 and USB. Software programs such as ATEasy allow the